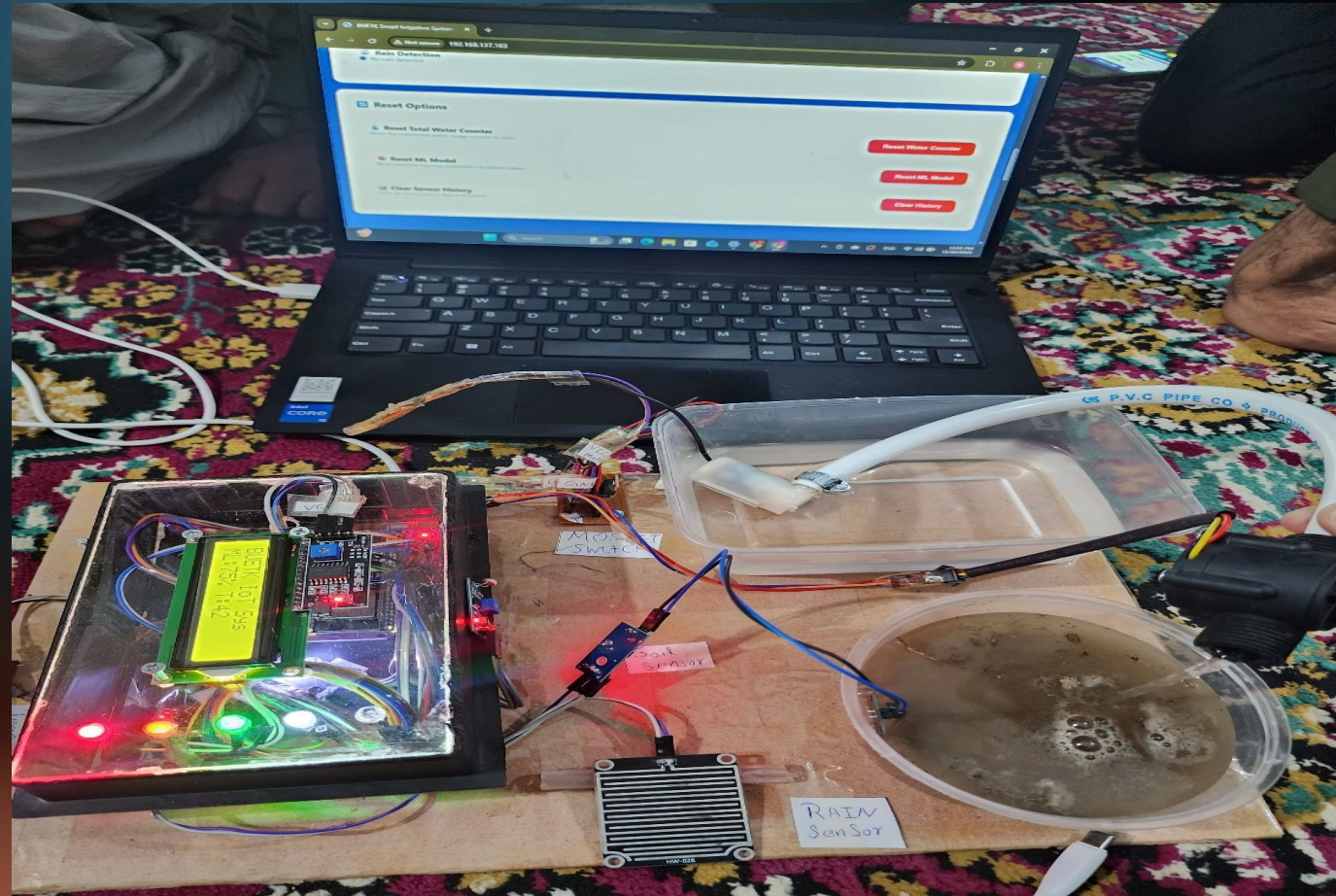




Smart Irrigation System Using Rule-Based Machine Learning and IoT For Water Management In BUETK

Presenter: Sajid Ali



Introduction

- Balochistan faces **severe water scarcity** yvaeh htiw , .thgilnus dna retaw llewerob no ecnailer
- Traditional irrigation is **manual, labor-intensive** dna , ot sdaels**significant water wastage**.
- Lack of automation makes conventional irrigation **inefficient and unreliable**.
- Modern solutions like **IoT** retaw evorpmi dna gnirotinom trams gnilbane yb tmemeganam .lortnoc
- IoT-based systems use sensors to collect **real-time soil and environmental data**.
- The **BUET-K Intelligent Irrigation System** seilppa dna To**rule-based machine learning** rof .noitamotua
- The system uses sensors and an ESP32 controller to decide **when and how much to irrigate**.
- It aims to achieve **water conservation ,sustainable agriculture**.natsihcolaB ni seussi noitagirri evlos dna ,



Problem Statement

- BUETK faces **severe water issues**.secitcarp noitagirri detadtuo dna ,retawdnuorg gninilced ,llafniar wol ot eud
- The current system uses **fixed timers**.sdeen tnalp lautca dna ,rehtaew ,erutsiom lios ekil srotcaf emit-laer gnirongi ,
- This results in **excessive water wastage** dna **uneven irrigation**.smraf hcraeser dna saera neerg ssorca
- A **smarter, automated, and environment-aware irrigation method**.deriuqer si
- The proposed project will use **IoT and rule-based machine learning** retaw etamotua dna atad emit-laer tcelloc ot KTEUB ta tnemeganam retaw elbaniatsus dna tneiciffe rof ylppus

Problem Objectives

- We can initiate accurate, data based irrigation that corresponds to crop requirements to reduce waste to enhance water-use efficiency.
- We can enable small and medium farmers better by providing affordable and easy-to-use smart irrigation solutions which can help them better utilize water and reduce their input expenses.

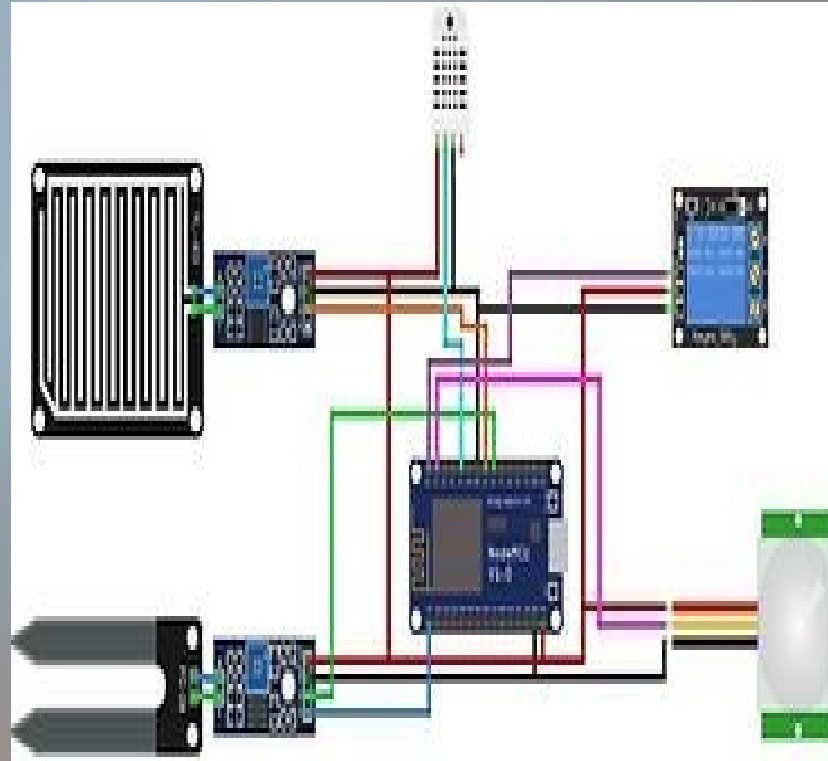
Problem Scope

- Develop an IoT-based smart irrigation system for BUETK using sensors (soil moisture, temperature, humidity, rainfall) and a microcontroller to monitor conditions in real time and automatically control water flow.
- Implement a rule-based machine learning model that uses sensor data to decide the correct irrigation schedule, helping reduce water wastage and improve efficient water management on campus.

Methodology

- Developed a Smart Irrigation System using IoT and rule-based machine learning to improve water management at BUETK.
- Selected key sensors: soil moisture, temperature, humidity, rainfall, and light.
- Connected all sensors to an ESP32 microcontroller to collect and transmit real-time data.
- Designed a simple rule-based ML logic to decide when and how much water the plants need.
- Automated the irrigation pump and valves based on the ML decisions for efficient water control.

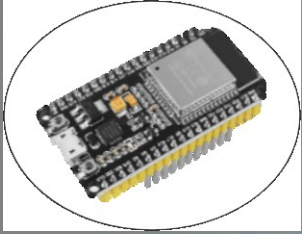
Circuit Diagram



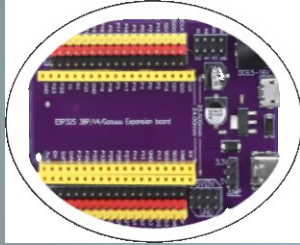
Block Diagram



Hardware



ESP32 MICROCONTROLLER



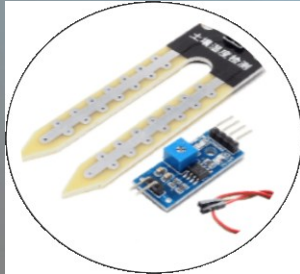
ESP32 SHIELD ADAPTER



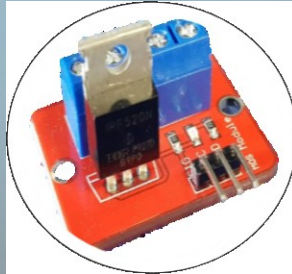
WATER FLOW SENSOR



LCD MODULE DISPLAY



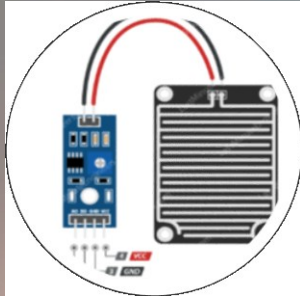
SOIL MOISTURE SENSOR



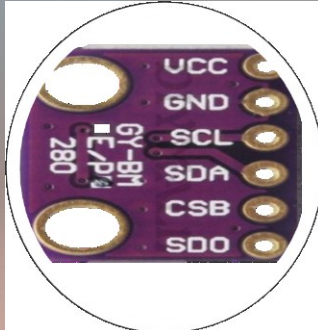
MOSFET SWITCH



SOLENOID VALVE



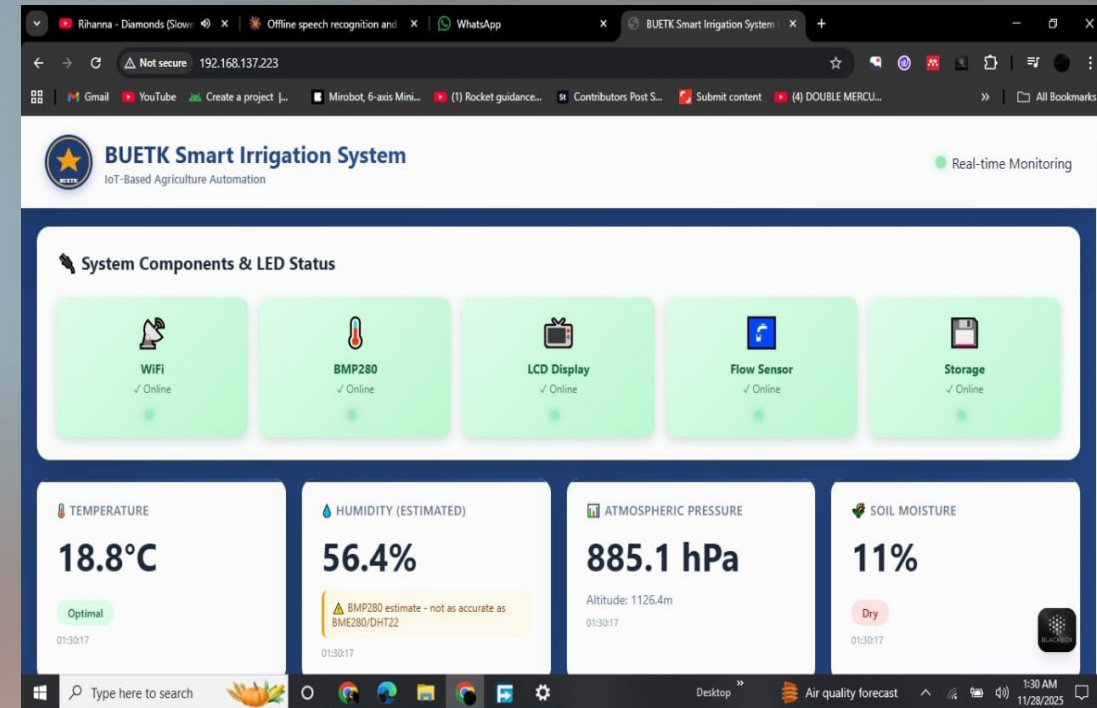
RAIN SENSOR



BME 280

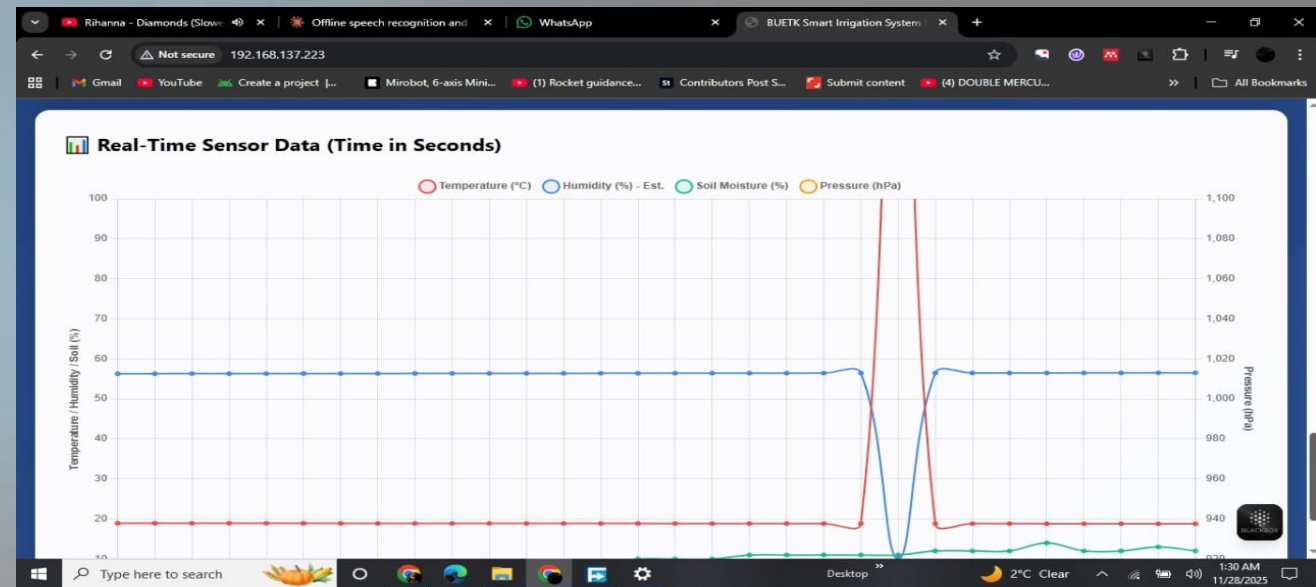
Software

WEB BASED DASHBOARD
SMART IRRIGATION SYSTEM USING RULED-
BASED MACHINE LEARNING AND IOT FOR
WATER MANAGEMENT IN BUETK



Result

- The graph shows time-series data for Temperature, Humidity, Soil Moisture, and Pressure recorded over a few minutes.
- Stable conditions:
- Pressure stayed around 1010 hPa.
- Humidity remained stable at .57%
- Soil moisture was very low, near baseline.
- Temperature remained around 20°C.
- Transient event:
- Temperature suddenly spiked from 20°C to ~100°C, then quickly returned to normal.
- Humidity dropped sharply during this event and then returned to .57%
- Soil moisture response:
- After the event, soil moisture increased slightly, indicating a minor or delayed watering effect.



Conclusion

- A Smart Irrigation System combining IoT and Rule-Based Machine Learning was successfully developed for efficient water management at BUETK.
- The system shifts irrigation from fixed-timers to precise, data-driven watering using real-time sensor readings.
- IoT sensors track soil moisture, temperature, humidity, and rainfall, enabling the ML model to decide when and how much to irrigate.
- This approach reduces water wastage, ensures plants receive the right amount of water, and supports sustainable resource use on campus.
- The solution is low-cost, scalable, and reliable, ideal for areas facing water scarcity like Balochistan.
- It not only enables automation and remote monitoring but also creates a foundation for advanced ML research and future development at the university.

Future work

- Upgrade the system from rule-based logic to advanced AI/ML models (e.g., LSTM, DNN) for predictive irrigation.
- Use historical soil moisture, weather forecasts, and evapotranspiration data to predict future water needs and move from reactive to proactive irrigation.
- Improve energy efficiency with optimized sensor network routing and long-term power management.
- Enhance scalability by using cloud platforms (e.g., ThingsBoard) and standard protocols like MQTT for centralized monitoring and multi-zone control.
- Add additional sensors (soil pH, nutrient levels) to support fertigation and complete crop management.
- Conduct long-term field trials on campus to measure water savings, crop improvement, and cost effectiveness.
- Assess and strengthen cybersecurity to protect IoT data and ensure secure system operation.

Reference

- Aqeel-ur-Rehman, A., Abbasi, A. Z., Islam, N., and Shaikh, Z. A. (2014).
Survey of the application of wireless sensors and networks in agriculture. *Computer Standards and Interfaces*, 36 (2) 263-270.
- Kumar, N., & Mallick, P. K. (2018).
The Internet of Things: The insights into the blocks, interactions, and layers of the components, and architecture. *Procedia Computer Science*, 132, 109-117.
- Jawad, H. M., Nordin, R., Gharghan, S. K., Jawad, A. M., and Ismail, M. (2017).
A review of energy efficient wireless sensor networks in precision agriculture. *Sensors*, 17(8), 1781.
- Patel, N. I., & Verma, H. K. (2020). Smart irrigation:
A review of the IoT based agriculture monitoring systems. *International Journal of Engineering Research and Technology*, 9 (4), 680-685.
- Machine learning-based smart irrigation system of IoT.
J Intelligent Fuzzy Systems, 40(2), 2341-2350.
- Food and Agriculture Organization (FAO). (2019).
Water shortage and arid area farming: Problem and solution. *FAO United Nations Report*.
- Balochistan integrated water resource management Report (IWRM). (2020).
- Ministry of water resources, Government of Pakistan.